Network inference in progressive MS stem cells: a robust stress-signalling programme, an unresolved cholesterol route

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1 Introduction

Network biology treats disease not as an isolated defect but as a perturbation that spreads through networks of interacting molecules [1]. Reconstructing such networks from omics data follows two main paradigms. Data-driven methods learn patterns directly from the measurements, for example co-expression networks [2] or factors shared across omics layers [3, 4], but they need large, sample-matched cohorts. Prior-knowledge network (PKN) methods instead start from a curated map of known interactions and ask which part of it best explains the changes measured in one condition. This step, called contextualisation, turns a generic map into a cell-specific mechanistic hypothesis and remains feasible with few samples [5].

PKN-based methods were developed separately for single biological layers. CARNIVAL fits signed signalling paths to transcription factor (TF) activities [6], whereas iMAT and flux balance analysis (FBA) fit metabolic flux under mass balance [7,8]. COSMOS joins signalling and metabolism, but does so by turning metabolic reactions into signed edges [9]. CORNETO [10] recently framed all of these as optimisation problems on a common hypergraph representation, which in principle allows them to be solved together. In practice, however, the methods are still combined at a late level: they are solved separately and compared or chained afterwards. A coupled signalling–flux model needs variables shared where the layers meet, for example an enzyme's activity in the signalling model bounding the flux of its reaction in the metabolic model, and such variables are not yet defined (Fig. 1c). The structure of the inferred network depends on the PKN's edges, on which nodes may start a signal and on how strongly small networks are preferred (Fig. 1b). How much of the inferred topology comes from the data, and how much from the structure of the network itself, hasn't been thoroughly tested.

Progressive multiple sclerosis (PMS) is a useful test case, because Ionescu et al. combined several omics layers with an explicit network hypothesis [11]. In induced neural stem cells (iNSCs) from people with PMS, they found a senescent, neurotoxic state that the cholesterol-synthesis inhibitor simvastatin reduced. They proposed a route from HMG-CoA reductase (HMGCR) through cholesterol to six cholesterol-responsive TFs (E2F1, EGR1, WT1, SP1, JUN and JUNB). Here we ask which TF activity changes distinguish PMS from control iNSCs, and whether contextualisation, without being told where to start, recovers the proposed route (Fig. 1a). We find a stress and senescence TF programme that is stable when any line is left out, connected to a kinase layer that recurs across both PKNs, but no support for synthesis-driven cholesterol signalling.

2 Methods

2.1 Data and measured nodes

We analysed public bulk RNA-seq of 3 control and 3 PMS iNSC lines (GSE297192 [12, 13]) together with metabolomics and lipidomics of 3 control and 4 PMS lines (Metabolomics Workbench ST003331/ST003332/ST003328 [11, 14]). Replicates were averaged, so that each cell line is one observation. Expression differences were scored with a moderated t statistic [15], and TF activities were inferred with CollecTRI regulons and decoupler [16, 17]. The 50 most changed TFs, all of which kept their sign when any one line was left out, became the measured signalling nodes. The joint TF–metabolite fit added 36 metabolite contrasts, including free cholesterol. The RNA-seq that Ionescu et al. analysed (2 vs 2 lines) comes from the same Park et al. study [12]. Because their raw proteomics are not public, their published table of 650 differentially abundant intracellular proteins (adjusted $p < 0.05$) served as an independent protein-level check.

2.2 Networks and contextualisation

The signalling PKN is OmniPath [18], restricted to signed, directed, curated edges between expressed genes (9,870 edges). The multi-omic PKN is COSMOS [9], which adds Recon3D metabolic reactions [19] and STITCH compound–protein interactions [20] (54,571 edges). CORNETO's CarnivalFlow selects the subnetwork that best reproduces the measured signs, with a penalty λ on the number of edges. Any node without incoming edges may start a signal, with a sign chosen by the solver. To avoid circularity, the nodes of the proposed route (HMGCR, SREBF1/2, SCAP, INSIG1, the six TFs, mevalonate and cholesterol) were never allowed to start one. We fitted TFs alone on both PKNs, and a joint TF–metabolite fit on COSMOS. Before any network was inspected, λ was set to the largest

value that kept the fitting error within 10% of the unpenalised ($\lambda = 0$) fit.

2.3 Testing what drives the inferred network

- *Is the route in the PKN?* We checked whether the reactions of the 20 cholesterol-synthesis enzymes (KEGG [21]) connect in pathway order, and searched for the shortest HMGCR → cholesterol path while penalising hub metabolites [22]. Pathway steps missing from COSMOS but present in Recon3D were restored in a repaired PKN.
- *Can the method find it at all?* As a positive control, we fixed HMGCR as inhibited, with the simvastatin-induced fall in cholesterol as the only measurement.
- *Data or network?* We refitted on ten degree- and sign-preserving randomised PKNs, under all 20 splits of the six RNA-seq lines into two groups, and with starting nodes restricted to receptors or with compound starting nodes removed.
- *Which nodes are favoured?* We regressed whether an unmeasured node was selected on its $\log_2$ PKN degree (logistic regression) in real and randomised networks, and ranked recurring nodes by degree and betweenness.
- *Does mass balance change the answer?* We built Recon3D (10,600 reactions) in CORNETO and fitted iMAT models [7] to the group-mean expression (top and bottom quartiles as high and low). Uptake was allowed for compounds detected in no-cell medium blanks, plus cholesterol, which the B27 supplement of the culture medium contains. FBA [8] then gave each model's capacity to make cholesterol with and without uptake.
- *Integration style.* We compared a metabolite-only fit merged afterwards with the TF-only fit (late integration) against the joint TF–metabolite fit on COSMOS, in which both layers are solved in one network problem (intermediate integration). Neither is early integration, which would combine matched samples before modelling: our layers come from different experiments.

Fits were solved as mixed-integer linear programs with HiGHS [23] in CORNETO 1.0.0rc8, for $\lambda = 0$, 0.01, 0.03, 0.1, 0.3 and 1, with a 1% optimality gap and 30–60 min time limits. Overlaps are Jaccard indices. Code, settings and input checksums are in the project repository.

3 Results

3.1 A stress TF programme, a recurring kinase layer, and HMGCR raised only as protein

PMS lines activated a stress and senescence programme: SP1 and JUN ranked second and sixth of 628 TFs and the AP-1 and NF-κB complexes 13th and 25th (Fig. 2a), while JUNB and EGR1 rose moderately, E2F1 was unchanged and WT1 is not expressed. In all three fits, the kinases MAPK1, MAPK14 (p38α), PRKACA, FYN and AKT1 sat directly upstream of these TFs (Fig. 4a), although the networks shared only 7–21% of their edges; all five are among the top 1% of OmniPath proteins by degree. Sterol-synthesis genes were not induced (Fig. 2b), but HMGCR protein was about nine-fold higher in PMS lines (adjusted $p = 2 \times 10^{-6}$), although protein and mRNA changes otherwise agreed (Spearman 0.51 over 606 genes).

3.2 The cholesterol route is in the network but is not selected

The synthesis pathway is in COSMOS except for one step (squalene synthase, FDFT1), which we restored in a repaired PKN. With HMGCR fixed as inhibited and the simvastatin-induced fall in cholesterol as the only measurement, CarnivalFlow recovered the canonical route in the repaired PKN but took a 32-edge mitochondrial detour in the original (Fig. 3a). From the PMS data alone, neither fit used synthesis: a single Golgi-to-cytosol transport step explained the cholesterol rise (Fig. 3b), and about 480 other metabolites are as close to SP1 as cholesterol is. The mass-balanced iMAT/FBA models agreed: control and PMS had the same active synthesis pathway, and when uptake was allowed neither required synthesis.

3.3 What the network contributes

The inferred networks were strongly sensitive to PKN structure and to the choice of starting nodes. First, *hubs are favoured*: each doubling of an unmeasured node's PKN degree raised its odds of selection 1.7- to 3-fold, to the same extent in randomised networks (Fig. 5c). Second, *a good fit is not evidence for the topology*: six of ten randomised PKNs fitted every TF sign exactly, while sharing

only 7–18% of inferred nodes with the real network. Third, *one starting node dominates, but which one is arbitrary*: a single starting node sat upstream of 24–92% of the explained measurements, and its identity changed with λ and PKN (Fig. 5a). In the joint TF–metabolite fit it was an unmeasured tamoxifen node, upstream of 39 of the 79 measurements that are reachable in the network (Fig. 4). Restricting starting nodes to receptors gave the same pattern, with one receptor (PTPRE) upstream of 49 of 50 TFs, and removing compound starting nodes moved the dominance to a single protein hub. The data nevertheless carry signal: of all 20 ways to split the six lines into two groups, only the true grouping and its mirror image were fitted without error ($p = 2/20$).

4 Discussion

The stress TF programme and its kinase layer fit the senescent phenotype: p38 MAPK and NF-κB drive the senescence-associated secretory phenotype [24, 25], and AP-1 organises the senescence transcriptional programme [26]. The raised HMGCR protein without transcriptional induction could reflect reduced INSIG-dependent degradation [27], which was not measured here. Every method in this study reads HMGCR only through its mRNA, so this possible entry point of the proposed route was invisible to them, and the network results neither confirm nor refute it.

More broadly, contextualisation could not settle the question because of four properties of PKN-based inference:

1. **Identifiability.** With thousands of candidate starting nodes, the edge penalty collapses the solution onto a single hub, and many hubs work equally well. The data constrain the last step into the TFs, not the path before it, which is why randomised networks fit as well as the real one.
2. **Hub bias.** Selection favours high-degree nodes even in randomised networks, so recurrence alone, as for the kinase layer, does not establish biological involvement.
3. **PKN completeness.** A single missing reaction (FDFT1) hid a textbook pathway, and nothing in the network signals such a gap.
4. **Signs are not chemistry.** In a signed network, an active enzyme "produces" its product without consuming anything, so one transport edge explains a rise in cholesterol as well as the whole synthesis pathway does. A stoichiometric model forbids this, because every molecule must be made or taken up, but without measured uptake rates it still cannot tell synthesis from uptake.

Integration styles. The Ionescu hypothesis predicts that a measured metabolite, cholesterol, lies upstream of the measured TFs, so we asked whether analysing the two layers together creates such links. In late integration, CarnivalFlow was run twice on COSMOS, once with only the TFs and once with only the metabolites, and the two networks were merged. They shared no node, so the merge could not link any metabolite to a TF. In the joint TF–metabolite fit, both sets of measurements entered one optimisation, in which an edge already used for one layer costs nothing extra when reused for the other. This re-routed the network (64 of 118 edges are new) but still placed no measured metabolite upstream of a TF; the layers met only through shared starting nodes, mainly the tamoxifen compound. CarnivalFlow and iMAT were likewise solved separately. A coupled signalling–flux model, in which an enzyme's inferred activity bounds its reaction's flux, would let the layers constrain each other; CORNETO makes this possible, but such shared variables are not yet defined.

Next steps. Four changes would make the upstream part of such networks testable. First, starting nodes should be limited to measured perturbations, for example simvastatin treatment or ligands detected in the secretome, instead of every source node in the PKN. Second, enzyme activity should enter at the protein level: the HMGCR increase seen here only in proteomics could then be a measured node rather than an inferred one. Third, CarnivalFlow and iMAT should be coupled in one CORNETO problem, so that an inactive enzyme forbids flux through its reaction and measured uptake and secretion from the spent medium bound the exchanges. Fourth, every inferred node should be compared with degree-matched null nodes, for example cholesterol against random metabolites of similar degree as a starting node, so that hub recurrence is not mistaken for evidence. Profiling the same cell lines in every layer would, in addition, allow factor models and PKN methods to be compared on equal terms.

Conclusion. The public PMS data support a stress and senescence TF programme and, at the protein level, a raised HMGCR. Network contextualisation cannot resolve whether HMGCR acts through cholesterol on these TFs: the step is invisible to mRNA-based inputs, and the upstream network is shaped by hubs and by the choice of starting nodes rather than by the data. Measured HMGCR activity, isotope tracing into cholesterol combined with a TF readout, or knockdown of the candidate TFs under simvastatin would distinguish the competing explanations.

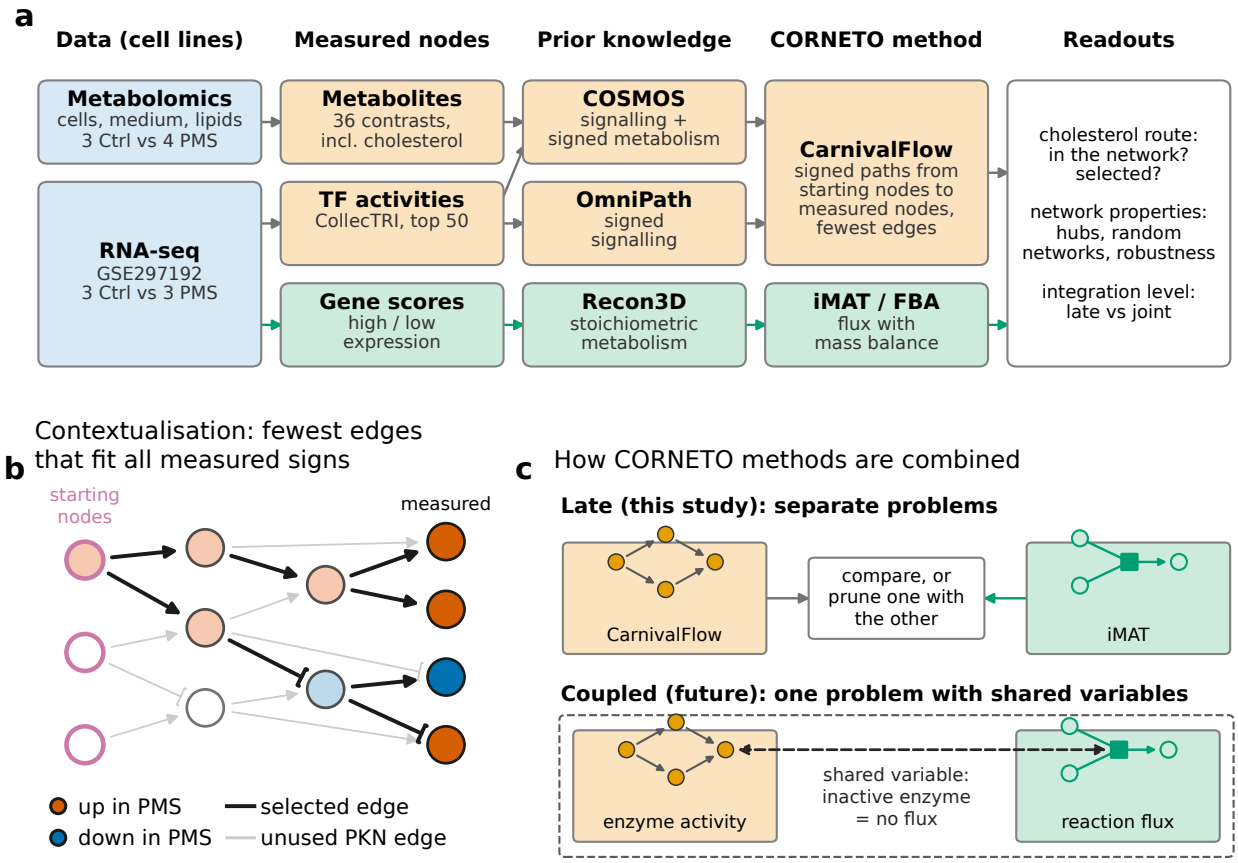


Figure 1: Data, networks and how the CORNETO methods were combined. (a) Workflow. Metabolomics and lipidomics: Metabolomics Workbench ST003331 (cells), ST003332 (medium) and ST003328 (lipids, with a simvastatin arm). The PKNs are OmniPath (9,870 edges), COSMOS (54,571 edges) and Recon3D (10,600 reactions). Orange: signed networks and CarnivalFlow; green: stoichiometric metabolism and iMAT/FBA. (b) Contextualisation, toy example. Signals may start at nodes without incoming edges (pink outline). CarnivalFlow keeps the fewest PKN edges whose signs reproduce every measured node; light colours are inferred states. (c) In this study the signed and stoichiometric methods were solved separately and combined afterwards (late). A coupled model would solve both in one problem, linked by shared variables such as an enzyme's activity bounding the flux of its reaction.

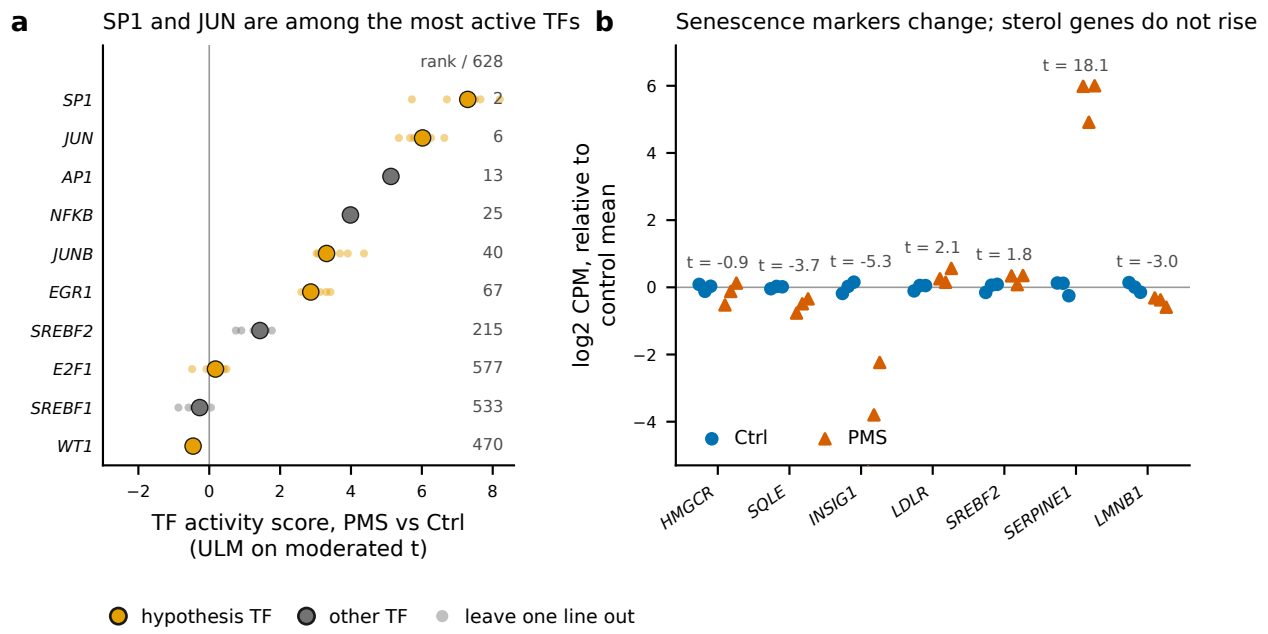


Figure 2: PMS lines show a stress TF programme without induction of sterol synthesis (3 vs 3 lines). (a) TF activity scores for the six hypothesis TFs (orange), the SREBFs and the AP-1/NF-κB complexes. Large dots: all lines; small dots: leaving out one line; numbers: rank among 628 TFs. Activity scores are inferred from each TF's target genes; WT1 itself is not expressed, so its score is not interpretable. (b) Per-line expression relative to the control mean, with the moderated t statistic. Senescence markers change, while sterol-synthesis genes do not rise.

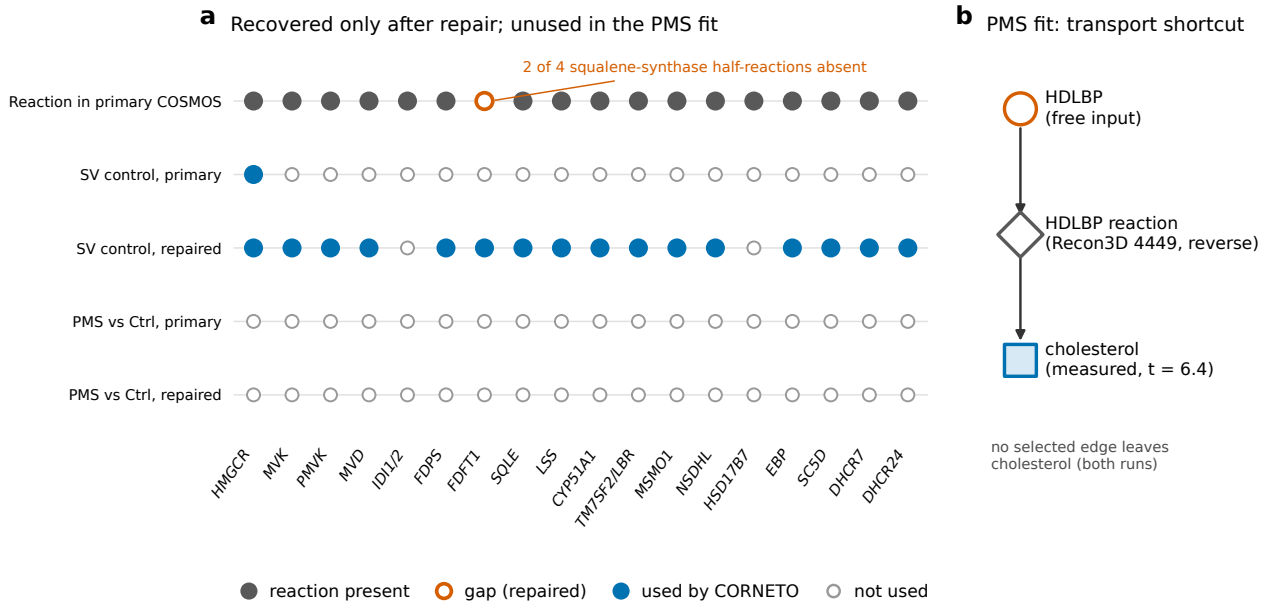


Figure 3: The cholesterol route is in the network but is used only for a known perturbation. (a) The cholesterol-synthesis enzymes in pathway order. Row 1: present in COSMOS; the two squalene-synthase (FDFT1) half-reactions are missing and were restored in the repaired network. Rows 2–3: enzymes used when CORNETO was told that HMGCR is inhibited and cholesterol falls (simvastatin), in the original and the repaired network; in the original network the solver reached cholesterol by a 32-edge detour through unrelated mitochondrial reactions. Rows 4–5: enzymes used in the PMS-versus-control fits. (b) In both PMS fits, the cholesterol rise is explained by one Golgi–cytosol transport step.

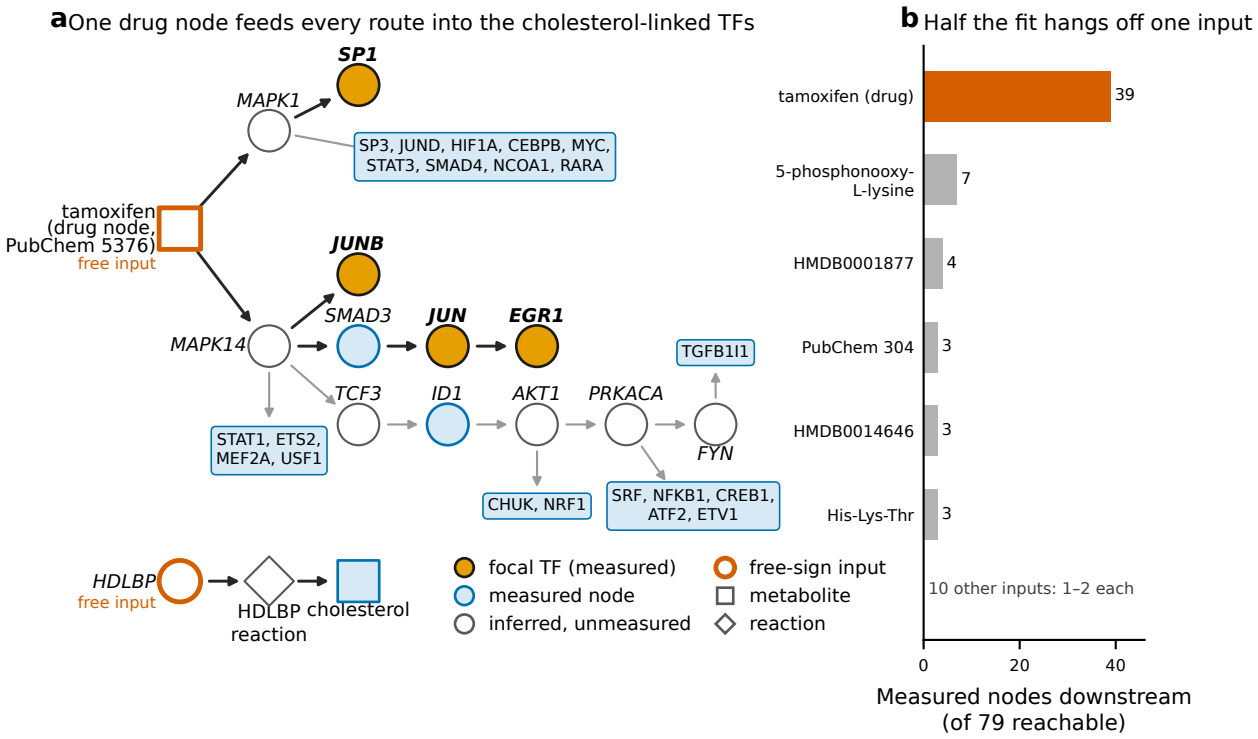


Figure 4: A kinase layer connects to the cholesterol-linked TFs in the joint TF–metabolite fit. The fit on COSMOS ($\lambda = 0.1$), restricted to the routes into SP1, JUNB, JUN and EGR1. (a) All routes pass through the kinases MAPK1 or MAPK14 and start at one unmeasured starting node, a tamoxifen compound; no route starts from cholesterol. Blue boxes collapse further measured TFs. All shown nodes are inferred as increased in PMS. (b) Measured nodes downstream of each starting node used, out of the 79 measurements reachable in the network (all 50 TFs and 29 of the 36 metabolites).

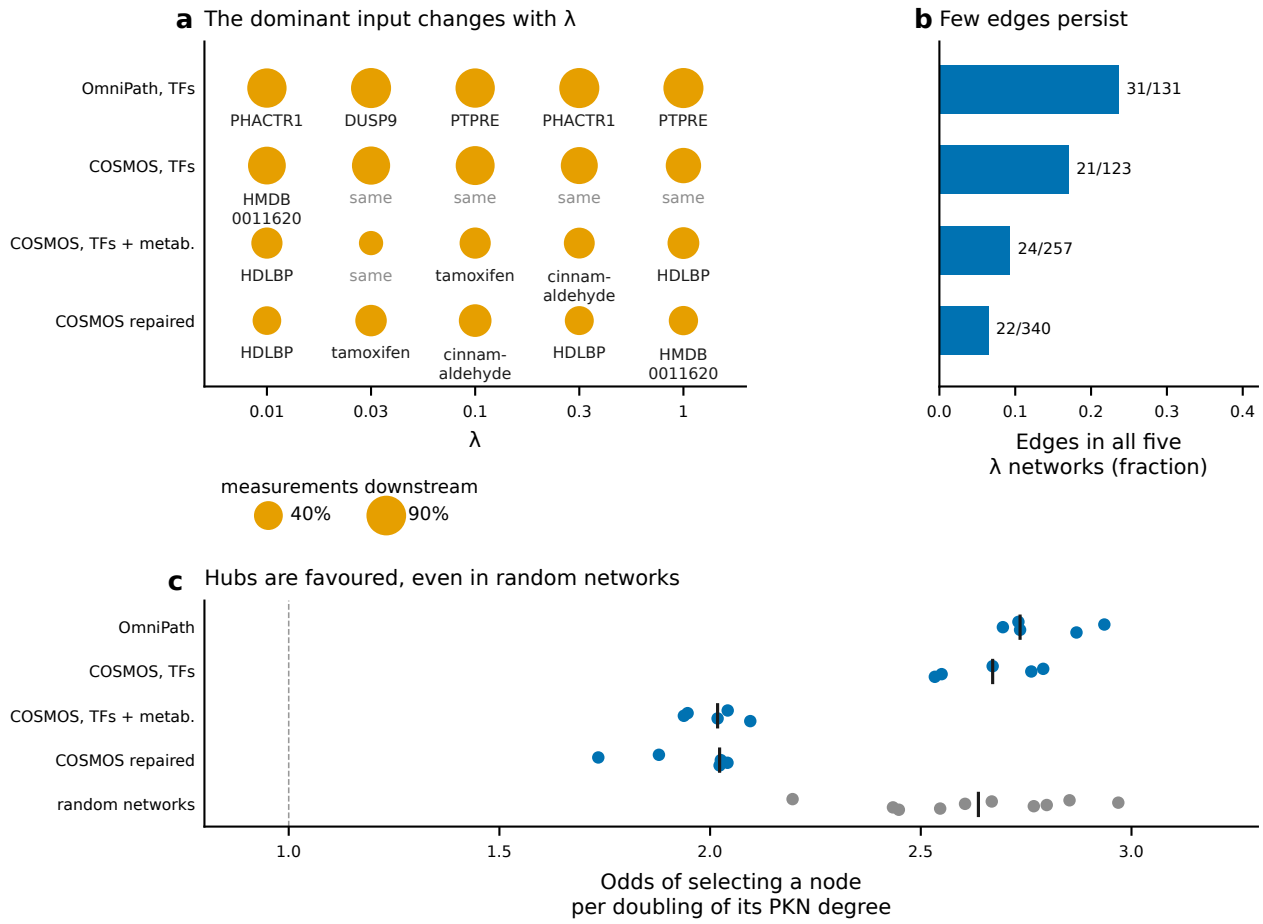


Figure 5: What changes, what persists, and why. (a) For each fit and penalty λ, the starting node upstream of the most measurements (dot area: fraction of measurements; "same": unchanged from the previous λ). (b) Edges selected at all five λ values, as a fraction of all edges ever selected. (c) Odds that an unmeasured node is selected, per doubling of its PKN degree (logistic regression; dots: λ values or random networks; bars: medians). Selection favours hubs to the same degree in degree-preserving random networks.

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